A detailed description of the first three steps of the proposed approach

Below are the details of the seven steps in our data-driven methodology. Preprocessed rs-fMRI data were temporally concatenated [1, 5] across subjects producing a *time-points* × *voxels matrix* (Y), as shown below:

$$\mathrm{Y} = \begin{bmatrix} \begin{bmatrix} y1_{1,1} & \cdots & y1_{1,v} \\ \vdots & \ddots & \vdots \\ y1_{tc,1} & \cdots & y1_{tc,v} \end{bmatrix} \\ \\ \vdots \\ \\ \begin{bmatrix} yn_{1,1} & \cdots & yn_{1,v} \\ \vdots & \ddots & \vdots \\ yn_{tc,1} & \cdots & yn_{tc,v} \end{bmatrix} \end{bmatrix}_{(n\times tc)\times vox}, \quad \mathrm{S} = \begin{bmatrix} SM^{ag}_{1,1} & \cdots & SM^{ag}_{1,v} \\ \vdots & \ddots & \vdots \\ SM^{ag}_{ic,1} & \cdots & SM^{ag}_{ic,v} \end{bmatrix}_{ic\times vox}$$

$$\rightarrow Y_{(n\times tc)\times vox} = A_{(n\times tc)\times ic} \times S_{ic\times vox} + \epsilon \tag{1}$$

Here, $n, tc, v(vox), ic$ refer to the number of subjects, time points, voxels, and independent components, respectively. Y is factorized as $Y = A.S + E$, where A, S and E represent the mixing matrix, spatial maps, and residuals. This decomposition captures temporal dynamics and spatial activity patterns by solving the following optimization problem:

$$min_{A,S}||Y - A.S||^2 + \lambda.Constraint(S, S_{template}) \tag{2}$$

where $S_{template}$ is the template, and λ controls the balance between fitting S to Y and adhering to the template. While the ideal model is $Y = A.S$, residuals inevitably arise from noise, model mismatch, and unexplained variability. The above expression is the objective function applied in CICA [8]. We used the NeuroMark template [3] for its reproducibility, subject-level flexibility, and biomarker-finding efficacy. In standard ICA (e.g., FastICA), no priors are used, and spatial maps are estimated solely from Y, yielding an objective of $min_{A,S}||Y - A.S||^2$. Y is decomposed based on principles from the Central Limit Theorem [7]. When independent data sources mix, the Central Limit Theorem [7] predicts the result will tend toward a Gaussian, as opposing signal fluctuations cancel out (destructive interference [11]), producing a mixture with reduced extremes [4, 9]. By 'unmixing' the data into a small set of Gaussian signals, ICA recovers the underlying independent sources—a core principle often referred to as blind ICA [2]. CICA [8], however, adds a constraint (with NeuroMark substituted into $S_{template}$) to guide the separation, improving precision and producing components that better align with known spatial brain patterns, and involves two stages: *forward estimation* and *inverse back-reconstruction*, as described below (see Figure 1 for more details). In forward estimation, S is first estimated by optimizing the objective function shown in the mathematical expression (2), and finally, A is estimated through the following process:

$$\mathrm{A} = \begin{bmatrix} \begin{bmatrix} y1_{1,1} & \cdots & y1_{1,v} \\ \vdots & \ddots & \vdots \\ y1_{tc,1} & \cdots & y1_{tc,v} \end{bmatrix} \\ \\ \vdots \\ \\ \begin{bmatrix} yn_{1,1} & \cdots & yn_{1,v} \\ \vdots & \ddots & \vdots \\ yn_{tc,1} & \cdots & yn_{tc,v} \end{bmatrix} \end{bmatrix}_{(n\times tc)\times vox} \times \left(\begin{bmatrix} SM^{ag}_{1,1} & \cdots & SM^{ag}_{1,v} \\ \vdots & \ddots & \vdots \\ SM^{ag}_{ic,1} & \cdots & SM^{ag}_{ic,v} \end{bmatrix}_{ic\times vox} \right)^{-1}$$

$$\rightarrow A = \begin{bmatrix} \begin{bmatrix} A1_{1,1} & \cdots & A1_{1,ic} \\ \vdots & \ddots & \vdots \\ A1_{tc,1} & \cdots & A1_{tc,ic} \end{bmatrix} \\ \\ \vdots \\ \\ \begin{bmatrix} An_{1,1} & \cdots & An_{1,ic} \\ \vdots & \ddots & \vdots \\ An_{tc,1} & \cdots & An_{tc,ic} \end{bmatrix} \end{bmatrix}_{(n\times tc)\times ic}$$

where the aggregated mixing matrix is obtained via $A = Y \times S^{-1}$ where each column of A A represents the concatenated time series of a specific brain activity pattern across all subjects. By inverting the individual mixing matrix for each subject and multiplying it by their respective original signal, we derive the individual spatial maps according to the following mathematical representation:

$$\rightarrow A = \left[\left[\begin{array}{ccc} & \cdots & \\ & \vdots & \\ & \cdots & \end{array}\right]\right]_{(n \times tc) \times ic}$$

$$A_i^{-1} \times Y_i = S_i$$

$$\rightarrow \begin{bmatrix} \\ \vdots \\ \end{bmatrix}_{ic \times tc} \times \begin{bmatrix} y_{i_{1,1}} & \cdots & y_{i_{1,v}} \\ \vdots & \vdots & \vdots \\ y_{i_{tc,1}} & \cdots & y_{i_{tc,v}} \end{bmatrix}_{tc \times vox}$$

$$= \begin{bmatrix} SM_{1,1} & \cdots & SM_{1,v} \\ \vdots & \vdots & \vdots \\ SM_{ic,1} & \cdots & SM_{ic,v} \end{bmatrix}_{ic \times vox}$$

where i denotes the subject index. Each row of S_i represents a unique spatial pattern indicating the voxel-wise contribution to brain activity, while each column of A_i corresponds to the time series of a specific component. Researchers prioritize the higher end of the low-frequency power range (0.01–0.15 Hz) when analyzing independent components, as this band reflects intrinsic baseline activity and contains the primary signal of interest for clinical investigations. Accordingly, each subject's S_i was aligned with NeuroMark's ICNs to quantify individual spatial contributions. Each ICN may consist of one or multiple components, meaning that components with similar patterns are grouped into the same ICN; this logic applies to the remaining ICNs across all domains. Step 2 involves computing pairwise correlations between the time series of all components for each subject over time. This results in a sequence of windowed matrices with dimensions of ICs $\times$ ICs, where ICs represents the total number of components across all ICNs. For example, if we have three ICNs consisting of 2, 3, and 1 components respectively, the resulting t windowed matrices would be 6×6. This sequence of windowed connectivity matrices constitutes dynamic functional network connectivity (dFNC), as illustrated in Figure 2.

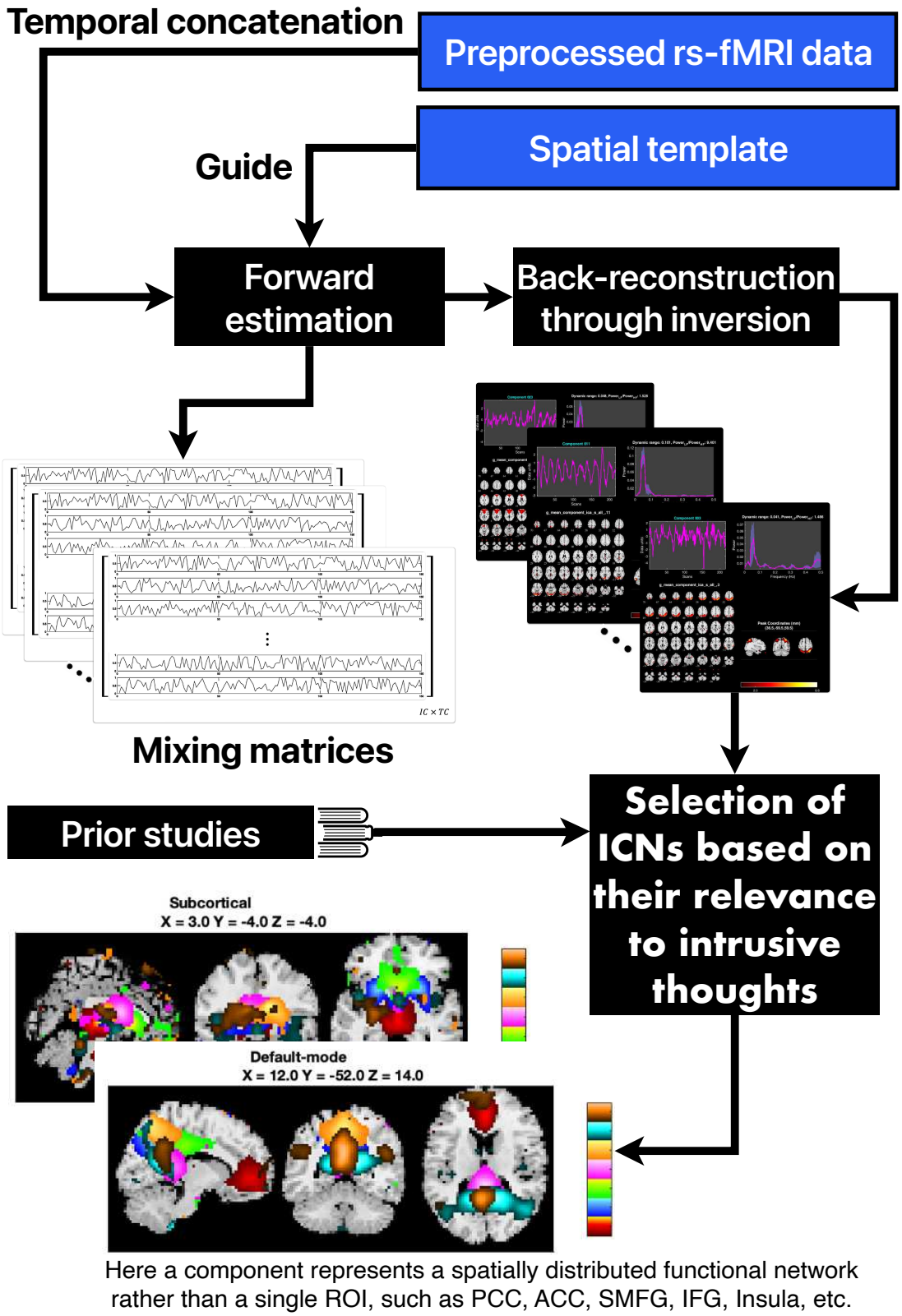


Figure 1: **Constrained Independent Component Analysis (Step 1).** Forward estimation stacks rs-fMRI data from all subjects together and unmix subjects one-by-one, guided by NeuroMark. Notably, there is no data leakage between subjects. This process results in a set of spatial maps accompanied by time courses. The number of extracted components depends on the user-defined parameter chosen at the beginning of the analysis, typically between 30 to 40 components are preferred. Using data from multiple subjects (i.e., longer time series or more spatial maps) helps reduce random noise in the estimation process, making the estimation of independence (non-Gaussianity, sparsity, etc.) more consistent. Thus, this relationship can be approximated as: $E = \frac{\sigma^2}{(N \times T)}$ where E represents noise that can affect accuracy, N is the number of subjects, T is the time course, and σ^2 is noise variance [6, 10]. For this reason, we included healthy controls in the estimation process but excluded them from subsequent analysis steps.

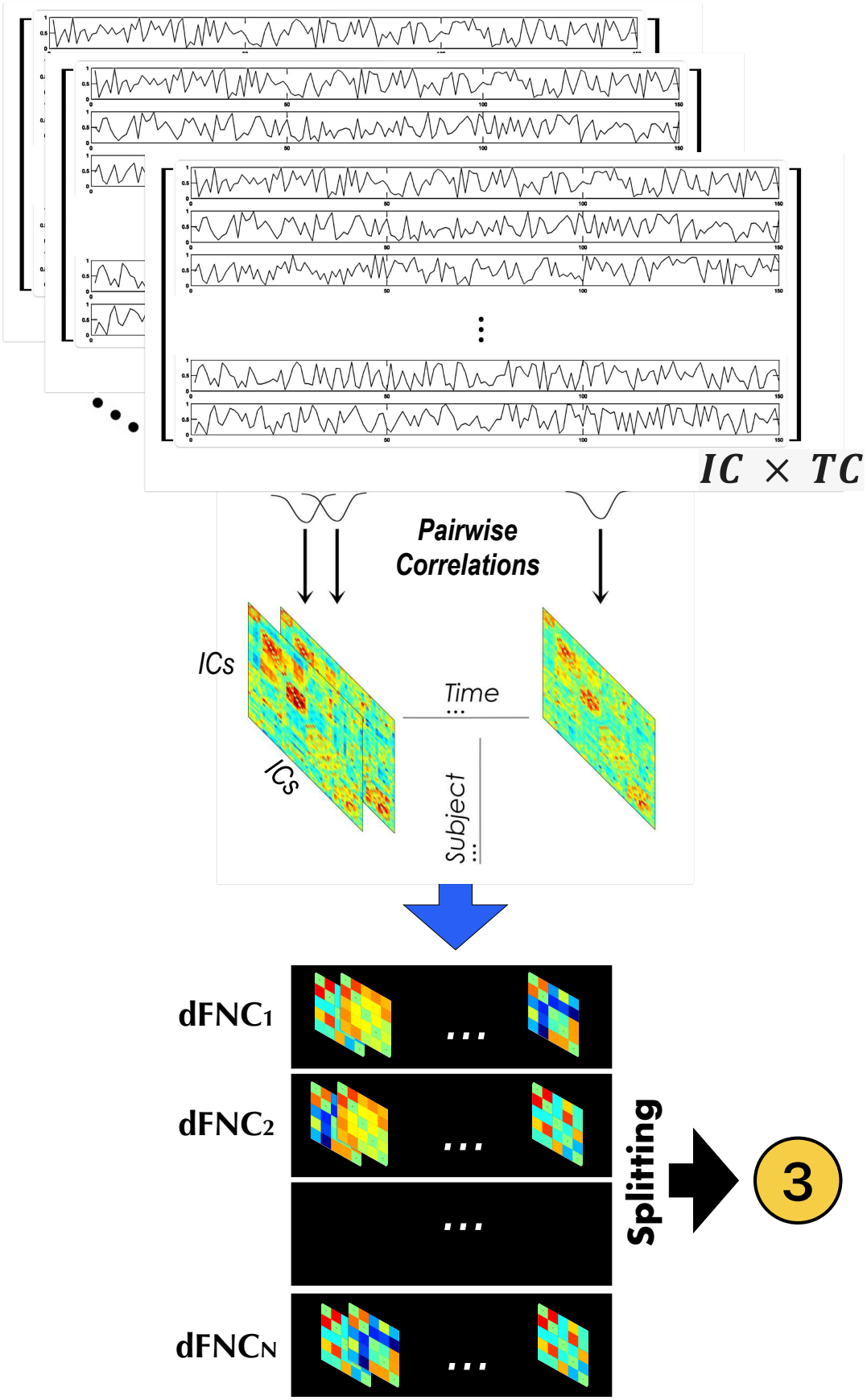


Figure 2: **Illustration of dFNC sequences across N subjects.** Matrices containing time series symbols are obtained from forward estimation of CICA (Figure 1). The time courses of the components of the selected ICNs—obtained from Step 1—are subjected to pairwise correlation of variable 1 and variable 2 across components, e.g., $IC-1$ and $IC-2$ for subject 1 correlated across time within window 1, and so on, using a sliding window algorithm to extract dynamic FNC matrices (dFNCs). The resulting dFNC sequences ($dFNC_1$ to $dFNC_n$) are then used for clustering, as illustrated in Figure 3.

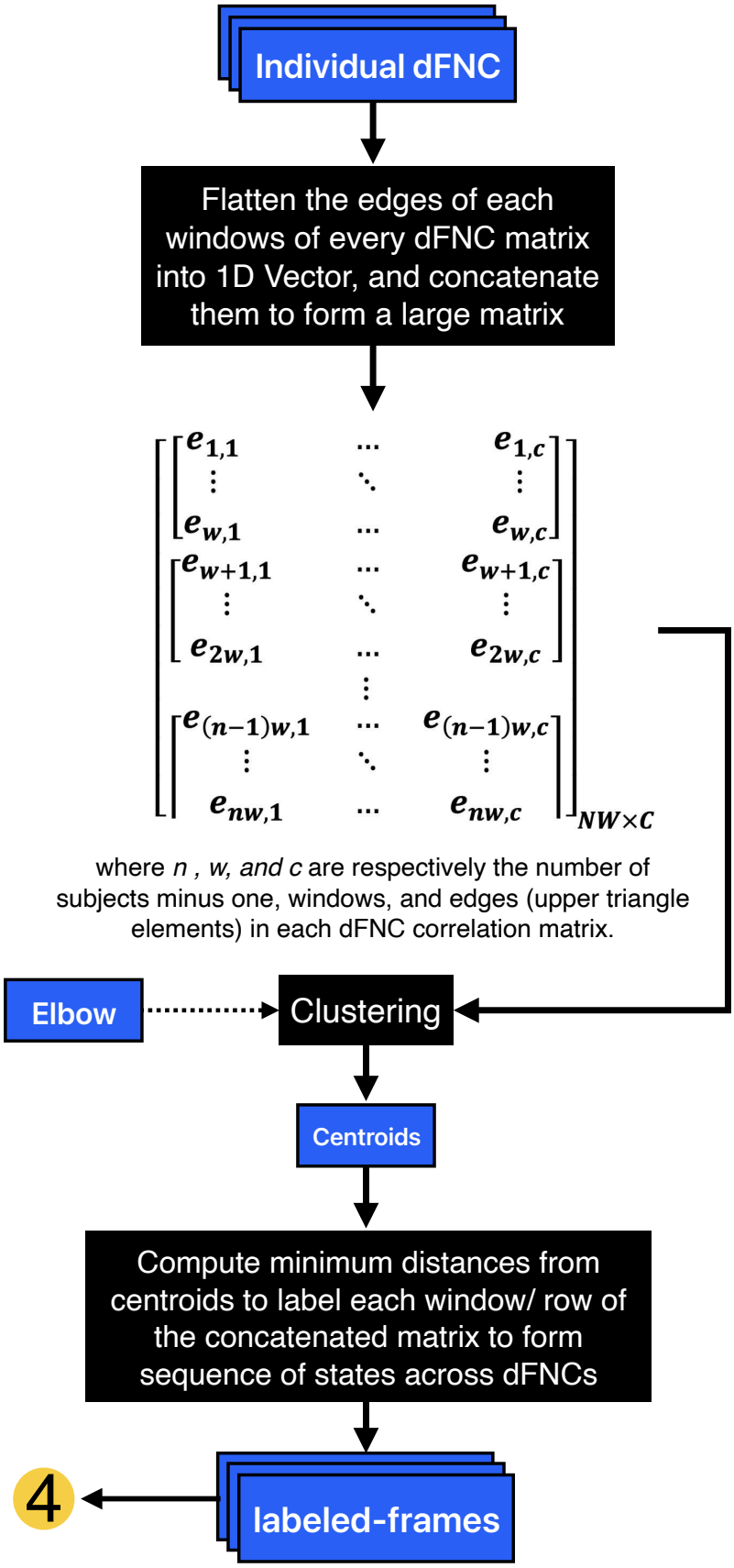
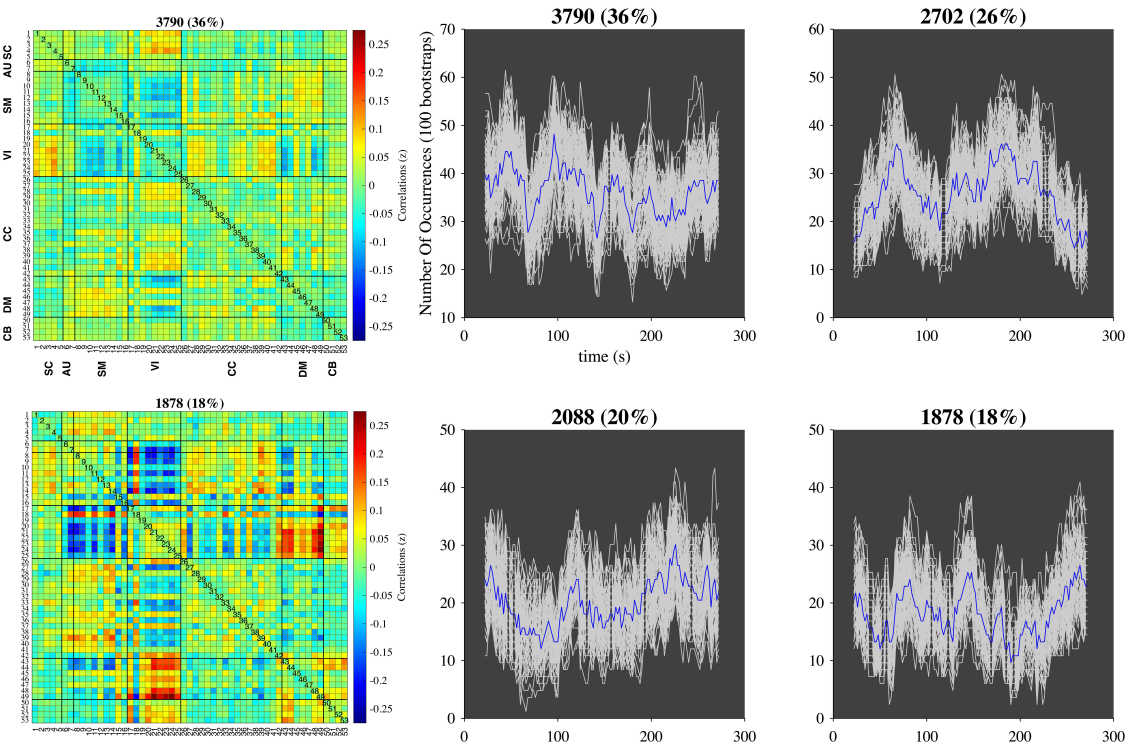


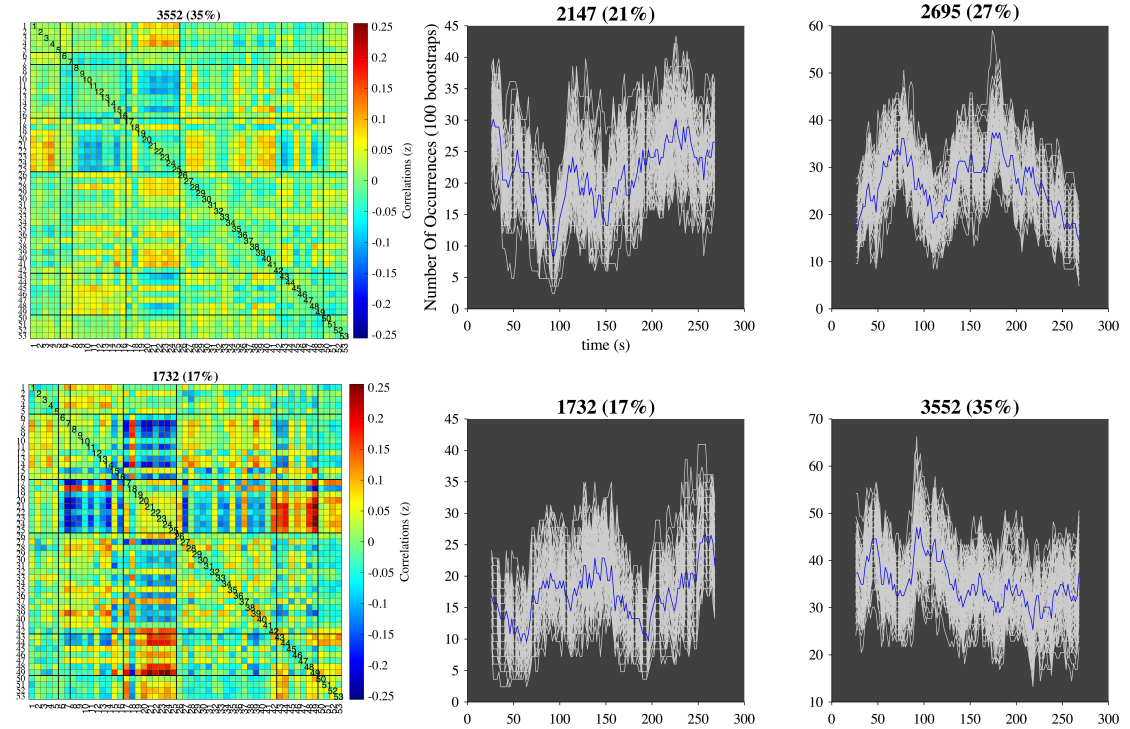
Figure 3: **Grouping windows and assigning labels (Step 3).** Each cluster centroid represented a distinct connectivity state, and the elbow criterion was used to determine the optimal number of states. In Step 4, frames were selected based on their assigned state labels.

Similarity of connectivity-state patterns across temporal scales

(a) 22–TRs



(b) 26–TRs



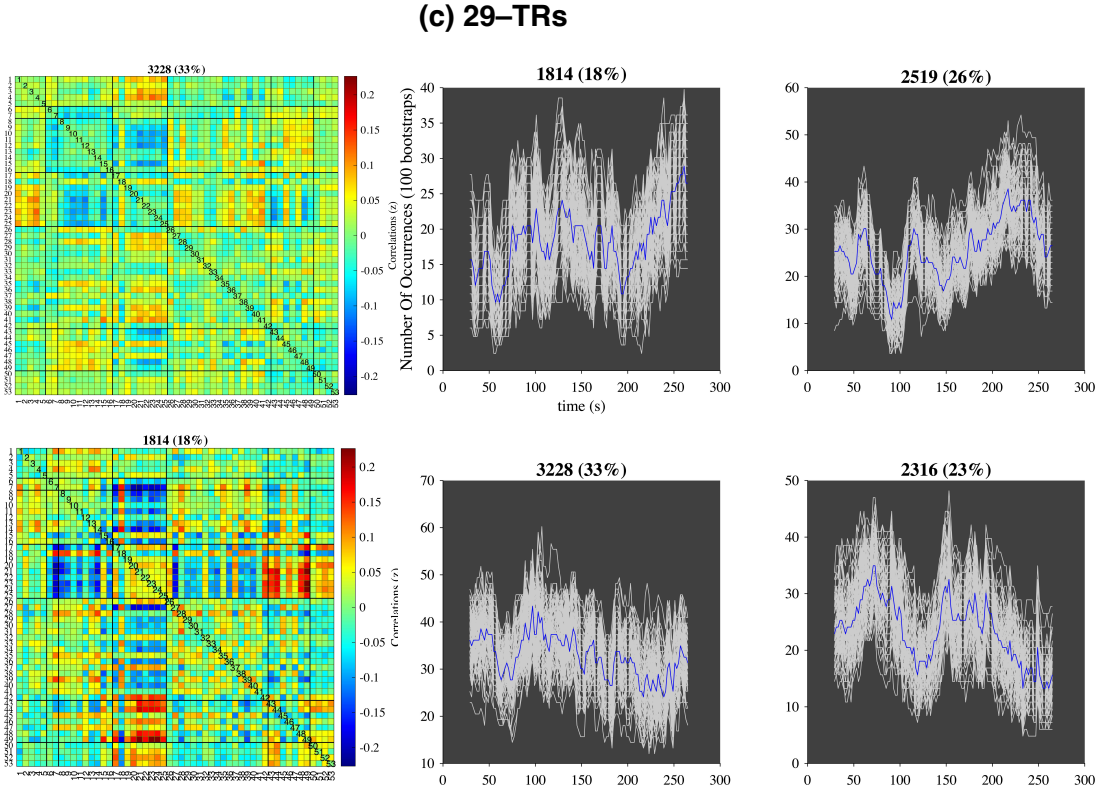


Figure 4: **Cluster centroids across temporal scales.** Connectivity patterns show similarities within and among the three most persistent states (States 1 in panel a, 4 in panel b, and 3 in panel c) and the three most transient states across the evaluated temporal scales.

References

[1] Vince D. Calhoun, Tülay Adali, Godfrey D. Pearlson, and James J. Pekar. A method for making group inferences from functional mri data using independent component analysis. *Human Brain Mapping*, 14(3):140–151, 2001.

[2] N. Correa, T. Adalı, and V. D. Calhoun. Performance of blind source separation algorithms for fmri analysis using a group ica method. *Magnetic Resonance Imaging*, 25(5):684–694, 2007.

[3] Y. Du, Z. Fu, J. Sui, S. Gao, Y. Xing, D. Lin, M. Salman, A. Abrol, M.A. Rahaman, J. Chen, and L.E. Hong. Neuromark: An automated and adaptive ica based pipeline to identify reproducible fmri markers of brain disorders. *NeuroImage: Clinical*, 28:102375, 2020.

[4] W. Fu. The theoretical basis of blind source separation. In *Principle and Application of Blind Source Separation Technology*, pages 19–40. Springer Nature Singapore, Singapore, 2025.

[5] Yundan Guo and Giuseppe Pagnoni. A unified framework for group independent component analysis for multi-subject fmri data. *NeuroImage*, 42(3):1078–1093, 2008.

[6] J. Himberg, A. Hyvärinen, and F. Esposito. Validating the independent components of neuroimaging time series via clustering and visualization. *NeuroImage*, 22(3):1214–1222, 2004.

[7] S. G. Kwak and J. H. Kim. Central limit theorem: The cornerstone of modern statistics. *Korean Journal of Anesthesiology*, 70(2):144, 2017.

[8] W. Lu and J.C. Rajapakse. Approach and applications of constrained ica. *IEEE transactions on neural networks*, 16(1):203–212, 2005.

[9] D. Lupu, J. L. Garrett, T. A. Johansen, M. Orlandic, and I. Necoara. Quick unsupervised hyperspectral dimensionality reduction for earth observation: A comparison. *IEEE Transactions on Computational Imaging*, 2025.

[10] M. J. McKeown, S. Makeig, G. G. Brown, T. P. Jung, S. S. Kindermann, A. J. Bell, and T. J. Sejnowski. Analysis of fmri data by blind separation into independent spatial components. *Human Brain Mapping*, 6(3):160–188, 1998.

[11] Jiangang Sun, Ling Liu, Guozhi Dong, and Jun Zhou. An extremely broad band metamaterial absorber based on destructive interference. *Optics Express*, 19(22):21155–21162, 2011.